ZCAS University

CCS2301 - OPERATING SYSTEM CONCEPTS

END OF SEMESTER EXAMINATION

5th June 2025

09:00 HRS - 12:00HRS

TIME ALLOWED: **WRITING – THREE HOURS**

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks.**
4. Numbering of your answers in your booklet should be the same as that in the question paper. Answers that are not numbered will **NOT be marked.**
5. Scientific Calculators **without** internet connectivity are allowed.
6. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

There are various algorithms which are used by the Operating System to schedule the processes on the processor in an efficient way. A Process Scheduler schedules different processes to be assigned to the CPU based on a particular scheduling algorithm. Popular process scheduling algorithms include the following two:

- Highest Response Ratio Next (HRRN) Scheduling (non-preemptive),
- Priority Based Scheduling (preemptive).

Requirements:

i) a) Given the Process IDs, the Arrival Time, and the Burst Time for each of the five (5) processes in table 1 below, you are required to determine the Completion Time, the Turn Around Time and the Waiting Time for each process using the Highest Response Ratio Next (HRRN) Scheduling algorithm (non-preemptive).

Note: *Copy the table into your answer booklet, clearly state the algorithm being used and fill in the 15 missing figures in the table. A table with figures but without a stated algorithm will NOT be marked.*

All the relevant calculations required to determine the next process to schedule MUST be clearly shown. Response Ratio is calculated by the given formula.

Response Ratio = (W + S) / S

Where, W → Waiting Time

S → Service Time or Burst Time

b) Calculate the Average Waiting Time **[Show the working]**.

Process ID	Arrival Time	Burst Time	Completion Time	Turn Around Time	Waiting Time
P1	0	4			
P2	2	6			
P3	4	5			
P4	6	2			
P5	8	3			

Table 1

(17 Marks)

ii) a) Given the Process IDs, the Priority, the Arrival Time, and the Burst Time for each of the seven (7) processes in table 2 below, you are required to determine the Completion Time, the Turn Around Time and the Waiting Time for each process using the Preemptive Priority Based Scheduling Algorithm.

Note: *Copy the table into your answer booklet, clearly state the algorithm being used and fill in the 21 missing figures in the table. A table with figures but without a stated algorithm will NOT be marked.*

In this table, the lower / smaller the number the higher the priority.

b) Calculate the Average Waiting Time [**Show the working**].

Process ID	Priority	Arrival Time	Burst Time	Completion Time	Turn Around Time	Waiting Time
P1	2	0	1			
P2	6	1	7			
P3	3	2	3			
P4	5	3	6			
P5	4	4	5			
P6	10	5	15			
P7	9	15	8			

Table 2

(23 Marks)

(Total: 40 Marks)

Section B

Question 2

Below are some of the basic linux operating system commands. You are required to state possible usage or options associated with each of the ten (10) listed commands below.

i) ls
ii) cp
iii) mv
iv) cd
v) mkdir
vi) rmdir
vii) cat
viii) grep
ix) ps
x) kill

(Total: 20 Marks)

Question 3

i) A "bit", a "byte", a "word", and a "block" are all units of data storage or data organization in a computer system. They are hierarchically related in terms of size, usage and structure. You are required to write brief notes to explain each of the four (4) terms and the relationships that exist among them. **(12 Marks)**

ii) The total time taken by swapping a process includes the time it takes to move the entire process from memory to a secondary disk and then copy the process back to primary memory. Let us assume that the standard hard disk where swapping will take place has a data transfer rate of around 1MB per second. How much time in milliseconds would an 8,192KB user process take to be swapped from the main memory to this standard hard disk and back to primary memory? [You should clearly show your working] **(8 Marks)**

(Total: 20 Marks)

Question 4

The following are some of the common Operating System Concepts and / or Terminologies that are associated with memory management. In **one sentence** or up to a maximum of **one paragraph**, you are required to provide an explanation of each of the ten (10) concepts / terminologies indicated below.

i) Virtual Memory
ii) Fragmentation
iii) Paging
iv) Segmentation
v) Translation Lookaside Buffer (TBL)
vi) Swapping
vii) Context Switching
viii) Demand Paging
ix) Pre-Paging
x) Deadlock

(Total: 20 Marks)

Question 5

The diagram below (Figure 1) summarizes various states that a process may change into during its life cycle. Using the diagram or otherwise, you are required to answer the questions below.

NOTE:

- Each question MUST be answered separately and CLEARLY numbered.
- NO explanations are required for questions (vi) to (x).
- BRIEF explanations are expected for questions (i) to (v).

i) Identify and explain any condition or situation that may cause the change of state for a particular process from suspend-wait state to waiting state.

ii) Identify and explain any condition or situation that may cause the change of state for a particular process from suspend-wait state to suspend-ready state.

iii) Identify and explain any condition or situation that may cause the change of state for a particular process from suspend-ready state to ready state.

iv) Identify and explain any condition or situation that may cause the change of state for a particular process from ready state to waiting state.

v) Identify and explain any condition or situation that may cause the change of state for a particular process from ready state to running state.

vi) Which part of a computer system do you expect to find a process that is in the running state?

vii) Which part of a computer system do you expect to find a process that is in the ready state?

viii) Which part of a computer system do you expect to find a process that is in the waiting state?

ix) Which part of a computer system do you expect to find a process that is in the suspend-wait state?

x) Which part of a computer system do you expect to find a process that is in the suspend-ready state?

(Total: 20 Marks)

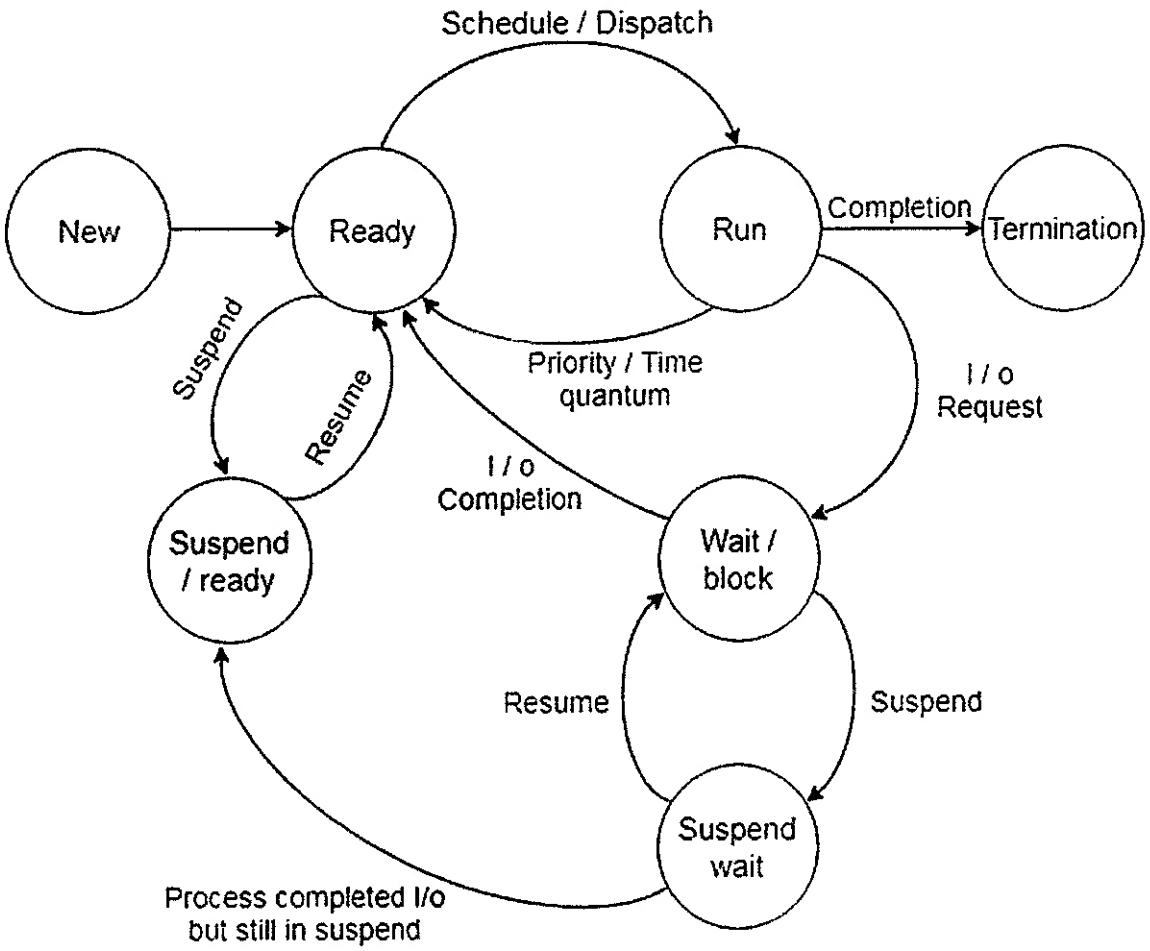


Figure 1

END OF EXAMINATION